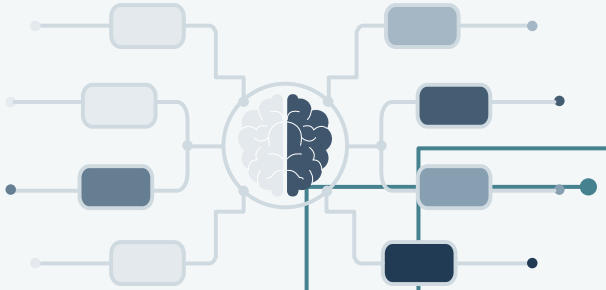
MIE 1624

Group 5

Proposal: Improving Canada's AI Strategy

Problem

- Global political crises disrupt economic stability
- Countries must prioritize **domestic technological development**
- Economic success increasingly tied to **innovation capacity**
- Focus: Canada's **AI innovation ecosystem**
- Goal: strengthen AI strategy to
  - Gain **competitive advantage**
  - Improve **overall AI capability**



Data Collection

Research & Innovation

Investment & Economy

Infrastructure & Compute

Talent & Workforce

Education & Academia

Adoption & Industry

Policy & Governance

Performing a Sentiment Analysis

Define Scope

Selected 40+ countries and identified AI-focused Keywords

Extract Data

Collected large-scale news data using NewsAPI

NLP Model

Used VADER to assign each article a sentiment score

Filter Data

Standardized text and ensured only AI-related articles were included.

Sum Up Insights

Computed average sentiment scores and compared AI perception across countries.

Visualize

Created visualizations showcasing the difference in perception towards AI

Random Forest vs XGBoost

Feature	Importance Score - SHAP
Private Investment ($B)	-0.38
citations	-0.19
Newly Founded AI Companies	-0.10
Average Gross	-0.10
Number of Universities in Top 100	-0.08
num_patent_applications	-0.04

Feature	Importance Score
accepted_patent_per_capita	0.310611
publications	0.153911
Average_Gross	0.124359
Median_Broadband__download_speed_MBPS	0.074728
total_data_centers_per_capita	0.062208
citations	0.059451

Implementation of AI Strategy

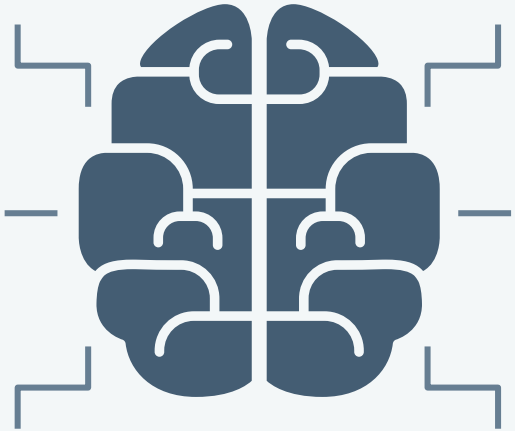
Increase Canada's AI Competitive-ness

- **AI Tax Credit** — Addresses lack of enterprise adoption
- **AI Research Internship** — Addresses talent and industry partnerships
- **AI Regulatory Sandbox** — Addresses enterprise sentiment

"AI-Adopt" Tax Credit

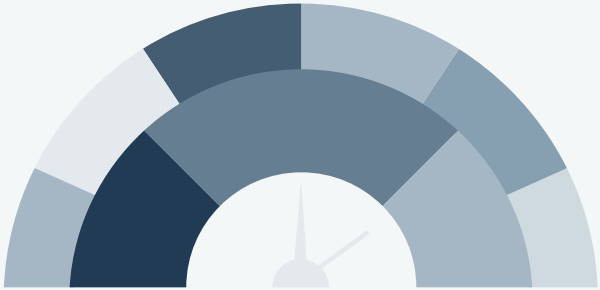
- Refundable **tax credit** for SME AI software licensing & digital consulting
- Singapore precedent: **AI adoption tripled in one year**; 85% reported productivity gains
- **Cost**: $500M over 3 years
- **Manager**: ISED Canada
- **Target**: 15% increase in SME AI usage

"Mitacs AI-Bridge" Internship



- **75% salary subsidy** for elite AI interns placed in manufacturing, agriculture & traditional sectors
- **Canada is #1** in G7 for AI talent growth — but **losing graduates to** ~$137K USD U.S. salaries
- **Cost**: $150M over 2 years | 1,000 placements/year
- **Manager**: Mitacs Research Canada
- **Target**: Each domestic AI job = ~$195K annual economic value (Vector Institute)

"Safe-Canada" Regulatory Sandbox



- Controlled testing **environment for AI startups** under future AIDA regulations
- Reduces regulatory friction → protects R&D pipeline behind patents & publications
- **Cost**: $75M over 2 years
- **Manager**: Treasury Board of Canada
- **Target**: 20% increase in VC funding for participating startups

Visualizing our findings and strategy

Canada AI competitiveness dashboard

Based on Tortoise Global AI Index 2024 · 43-country dataset

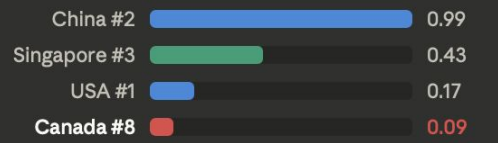
Rank #8 of 43 · Tortoise 2024

① What Canada is lacking | ② Why it matters | ③ What changes will do

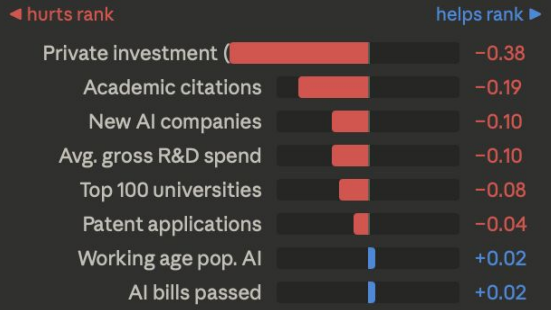
AI adoption rate	Public sentiment	Broadband speed	AI bills passed
48.1%	0.088	256 Mbps	0
−9.4pp vs SG	−0.90 vs CN	−151 vs SG	No AIDA yet

AI adoption vs. top nations

Public sentiment score



SHAP drivers — Canada's rank penalty



Values = SHAP contributions to log-rank. Model Spearman ρ = 0.989, R^2 = 0.96.

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① What Canada is lacking | ② Why it matters | ③ What changes will do

Value per AI job	Mitacs productivity lift	US grad median wage	Sandbox funding lift
$195K	+11%	$137K USD	+15%
annual GDP contribution	vs. national average	Brain drain risk	capital raised (UK FCA)

Why private investment is the #1 gap

Private investment carries the single largest negative SHAP weight (−0.38). Canada's $15B (2013–2024) lags far behind the US and China. The model shows every additional billion in private AI investment has an outsized positive effect on rank.

US/CN figures illustrative — Canada figure from dataset.

Canada's strengths worth protecting

- **6 universities in global CS top 100**
  Toronto, McGill, UBC, Waterloo and more
- **492 newly founded AI companies (2013–2024)**
  Strong base — needs scale-up capital
- **2.2M+ developers**
  #1 G7 for year-over-year AI talent density growth
- **Legislative momentum building**
  AIDA framework in progress — 1 AI mention in proceedings

Comparative benchmark — Canada vs. elite 3

Country	Rank	AI usage	Sentiment	Broadband	Priv. invest.
USA	#1	51.5%	0.17	303 Mbps	~$335B
China	#2	54.3%	0.99	223 Mbps	~$175B
Singapore	#3	57.5%	0.43	407 Mbps	—
Canada	#8	48.1%	0.09	256 Mbps	$15B

Canada AI competitiveness dashboard

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① What Canada is lacking | ② Why it matters | ③ What changes will do

Current rank

#8

Tortoise Global AI Index 2024

→ If all 3 strategies are implemented

Target rank

Top 5

Within 3 years

AI-Adopt tax credit

Refundable federal credit for SMEs adopting Canadian-made AI tools. Modeled on Singapore's SMEs Go Digital program — tripled adoption in one year, 85% of firms reported productivity gains.

$500M · 3 years · ISED Canada

Target: +15pp SME AI usage

Mitacs AI-Bridge internship

75% salary subsidy for elite graduates placed in manufacturing and agriculture. Combats brain drain — STEM grads significantly more likely to migrate. Each retained AI job generates $195K in economic value annually.

$150M · 2 years · Mitacs + ISED

Target: 1,000 elite placements/yr

Safe-Canada sandbox + AIDA

Regulatory sandbox for AI startups to test under AIDA compliance. UK FCA equivalent: firms see 15% more capital raised and 50% higher funding probability. Builds institutional trust and closes sentiment gap.

$75M · 2 years · Treasury Board

Target: +20% VC funding for AI

Projected impact by indicator

SME AI adoption rate	48.1% → ~63%	+15pp
Public sentiment score	0.09 → ~0.30	+0.21
VC funding for AI startups	baseline → +20%	sandbox effect
Elite grad retention (domestic)	low → 1,000/yr kept	brain drain reversed
AI bills passed (legislative clarity)	0 → AIDA passed	regulatory certainty

Total estimated investment **$725M over 3 years**

What now?

Students

Build AI Projects, research, file patents , contribute to open source

Employers

Invest in AI adoption, pilot AI in operations, support employee training

Employees

Use AI in your daily work, automate repetitive tasks, continue to learn

Government

Fun AI programs and startups, Incentivise adoption(tax credits for example), strengthen AI policy and regulations

Thank You

Please feel free to ask any questions.

Our Chatbot

🧠 What is it Based On?

- **Structured Data**
  - AI sentiment dataset (db_merged.csv)
  - Country-level summaries
- **Project Knowledge**
  - AI Index frameworks (Stanford, Tortoise)
  - Identified weaknesses & recommendations

⚙️ How It Works

User Question → Retrieve Data → Prompt Construction → LLM (Gemini API) → Answer

- Built with:
  - Python (Pandas)
  - Gemini 3.1-pro preview API (LLM)
  - Gradio (UI)

📦 Architecture

Multi-agent LLM chatbot using **RAG (Retrieval-Augmented Generation)**

- **Planner** decides how to handle the query
- **Researcher** retrieves relevant documents & reports
- **Analyst** extracts insights from structured datasets (e.g., sentiment tables)
- Combine retrieved context + data → construct prompt